Percutaneous Drainage for Postoperative Fluid Collection after Hepatobiliary Pancreatic Surgery

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Abbreviation: PFC = postoperative fluid collection

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Postoperative fluid collection (PFC) frequently occurs after major pancreatic resection and hepatobiliary surgery. Percutaneous drainage is used as the first-line treatment because of the high mortality rate of reoperation for PFC. However, various complications have been reported with use of percutaneous drainage, despite the high success rate. Drainage failure is associated with a higher mortality rate, whereas successful drainage is key to improving the postoperative prognosis. Safe puncture of the PFC after hepatobiliary pancreatic surgery is difficult because the deep lesions occur in anatomically complex locations surrounded by organs and vessels (Fig 1). In the online presentation, the authors provide current knowledge and techniques to enable safe drainage at these sites. Treatment concepts and strategies include evaluation of the PFC, puncture and catheter placement, catheter introduction, and postdrainage management.

Evaluation of the PFC is essential because the indication for drainage must be based on an accurate diagnosis. Confirming the CT findings of the PFC is critical, in addition to the clinical findings, to determine the internal structure, size, and shape of the abscess.

Puncture and catheter placement are the most important parts of the drainage procedure and require selection of an imaging guidance modality, puncture route, puncture method, puncture kit, and drainage catheter. First, the most appropropriate modality for imaging guidance (US, CT, CT fluoroscopy, or fluoroscopy) and the puncture route are selected (Fig 2). The optimal puncture method (trocar technique or Seldinger technique) is selected to perform percutaneous drainage. Selecting an appropriate imaging guidance modality and determining a safe and effective puncture route are vital.

Catheter introduction is key to reduce the abscess cavity size in cases of residual abscess. Catheter introduction requires confirming the abscess of origin of the PFC before catheter introduction, understanding the catheter introduction system, and recognizing the difference between the safety and selective wires. It is important to understand that hydrophilic wire usually has high selectivity, and spring wire with an appropriate stiffness serves as a safety wire.

Timing the removal of the drain appropriately to prevent abscess recurrence is of paramount importance during postdrainage management. Before the tubing is removed, the catheter removal criteria should be checked, and if the criteria are met, the catheter can be

TEACHING POINTS

- Safe puncture of the PFC after hepatobiliary pancreatic surgery is difficult because the deep lesions occur in anatomically complex locations surrounded by organs and vessels.
- Knowledge of devices, techniques, puncture routes, and catheter introduction allows safe drainage after hepatobiliary pancreatic surgery.
- Advanced puncture and catheter introduction techniques can help increase the effectiveness of drainage after hepatobiliary pancreatic surgery.

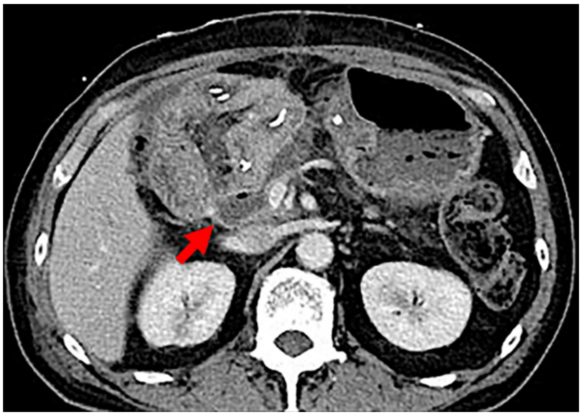

Figure 1. Axial contrast-enhanced CT image obtained after pancreaticoduodenectomy shows a typical deep PFC (arrow) near the portal vein.

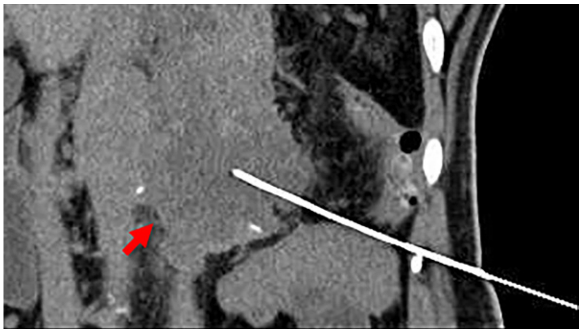

Figure 2. Coronal maximum intensity projection CT image was obtained after puncture for a typical deep PFC near the pancreas stump (arrow) under US guidance. The needle tip reached the target abscess and avoided the intestine.

removed. Sinography should be performed to assess for the presence of an internal fistula in cases of anastomotic leakage before tube removal.

Advanced puncture and catheter introduction techniques based on an understanding of therapeutic concepts and strategies for drainage include the right intercostal approach, advanced puncture techniques (changing needle direction technique or bent-needle technique), percutaneous drainage with preexisting surgical drains, catheter tract formation and catheter tract puncture technique, and a navigation map for catheter introduction or puncture. These techniques are needed to safely puncture deep lesions in anatomically complex locations surrounded by organs and vessels after hepatobiliary pancreatic surgery.

The online presentation reviews the basic knowledge and appropriate treatment strategies for percutaneous drainage of PFCs and describes fundamental techniques. Drainage should be based on accurate knowledge, appropriate treatment strategies, and reliable techniques to avoid failed drainages or complications that could worsen the postoperative prognosis.

Suggested Readings

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